THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS FULL SYLLABUS PAPER SET 03

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A complete graph with 4 points has how many lines joining them? 1

(a) 12 **(b)** 6 **(c)** 4 **(d)** 8

2. The sum of the first 20 odd numbers is: 1

(a) 400 **(b)** 40 **(c)** 420 **(d)** 380

3. A line extends: 1

(a) endlessly in both directions **(b)** only between two points

(c) in one direction only **(d)** in a closed curve

4. The unit of angle measurement is: 1

(a) litre **(b)** centimetre **(c)** degree **(d)** gram

5. The estimate of 498 + 303 by rounding to hundreds is: 1

(a) 900 **(b)** 1000 **(c)** 700 **(d)** 800

6. The digits of a number may be rearranged to make the largest number by writing them: 1

(a) in descending order **(b)** alphabetically

(c) randomly **(d)** in ascending order

7. A bar graph has bars of heights 7, 9, 14, 10. The total is: 1

(a) 40 **(b)** 37 **(c)** 42 **(d)** 45

8. In a bar graph the bars have: 1

(a) equal width and equal gaps **(b)** different widths

(c) no gaps ever **(d)** curved tops

9. A number is divisible by 9 if: 1

(a) it ends in 9 **(b)** it has two 9s

(c) it is odd **(d)** the sum of its digits is divisible by 9

10. Which of these is a prime number? 1

(a) 27 **(b)** 39 **(c)** 33 **(d)** 29

11. The area of a rectangle of perimeter 20 cm and sides 6 cm and 4 cm is: 1

(a) 10 cm² **(b)** 20 cm² **(c)** 48 cm² **(d)** 24 cm²

12. Perimeter of a rectangle 8 cm × 5 cm is: 1

(a) 40 cm **(b)** 26 cm **(c)** 13 cm **(d)** 21 cm

13. Fractions with the same denominator are called: 1

(a) like fractions **(b)** unlike fractions **(c)** mixed fractions **(d)** unit fractions

14. $\frac{3}{4} - \frac{1}{2}$ equals: 1

(a) $\frac{1}{4}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{1}{8}$ **(d)** $\frac{3}{8}$

15. A rectangle of sides 18 cm and 10 cm has perimeter: 1

(a) 180 cm **(b)** 28 cm **(c)** 58 cm **(d)** 56 cm

16. The diagonals of a square are: 1

(a) equal and perpendicular **(b)** parallel
(c) of length zero **(d)** unequal

17. The Indian flag's Ashoka Chakra (24 spokes) has an order of rotational symmetry of: 1

(a) 24 **(b)** 6 **(c)** 12 **(d)** 8

18. The order of rotational symmetry of a regular pentagon is: 1

(a) 2 **(b)** 5 **(c)** 1 **(d)** 10

19. A lift moves from the ground floor to 3 floors below. The integer is: 1

(a) −1 **(b)** 0 **(c)** 3 **(d)** −3

20. In the token model, a red token represents: 1

(a) 0 **(b)** 2 **(c)** −1 **(d)** +1

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 10–15 words each.

21. State the rule of the sequence 2, 5, 8, 11, … and give the next term. 2

22. Why is a number ending in 0 or 5 divisible by 5? 2

23. From a pictograph, one symbol = 8 students. If a class has 5 symbols, how many students? 2

24. Is 2,134 divisible by 3? Give reason. 2

25. Do two figures with equal perimeters always have equal areas? Give an example. 2

26. Add $\frac{1}{2} + \frac{1}{4}$. 2

27. What do we need to construct a square? 2

28. A diver is at −12 m and goes 5 m up. Where is she now? 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Hexagonal numbers are 1, 7, 19, 37. Find the differences between consecutive ones and describe the pattern. 3

30. Name the types of angle formed by the hands of a clock at 1:00, 3:00 and 5:00. 3

31. How can data be misrepresented in a graph? Give an example. 3

32. A parallelogram of base 10 cm and height 6 cm is cut and rearranged into a rectangle. Find its area. 3

33. Draw a square with diagonal 6 cm using properties of diagonals. 3

34. State which has greater order: a regular pentagon or a regular hexagon. Why? 3

35. Explain: every even number greater than 2 is composite. 3

OR

Why is the divisibility rule of 3 true for 123? Verify. 3

36. Brahmagupta gave rules about debts and fortunes. Express: a debt of 5 added to a debt of 3. 3

OR

Write the integers between −4 and 3 and find their sum. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 80–100 words each.

37. (i) State the sum of angles at a point. 5
(ii) If three angles at a point are 100°, 120° and x, find x.
(iii) Name the type of x.

OR

(i) Draw angles of 40°, 90° and 150°. 5
(ii) Classify them.
(iii) Arrange them in descending order.

38. (i) How is the number line used for comparing numbers? 5
(ii) Order 4,508; 4,058; 4,805; 4,850.

OR

(i) What is a number line? 5
(ii) Mark 0, 250, 500, 750 and 1000 on a line.
(iii) Which of these is nearest to 640?

39. (i) State why the denominator cannot be zero. 5
(ii) Write the smallest unit fraction with denominator ≤ 10.
(iii) Order $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{10}$.

OR

Case study: A tailor has 5 m of cloth. A shirt uses $1\frac{3}{4}$ m. 5
(i) How much for two shirts?
(ii) How much cloth is left?
(iii) Can he make a third shirt?
(iv) How much more is needed or left?

40. **Case study:** A tailor cuts a paper folded in half to create a heart. 5
(i) What symmetry does the cut-out have?
(ii) How many lines?
(iii) Does it have rotational symmetry?
(iv) Give another example.

OR

Case study: A logo is a regular pentagon with a star inside. 5
(i) Lines of symmetry?
(ii) Angle of rotation?
(iii) Does it match when rotated by 72°?
(iv) Draw it.

END OF QUESTION PAPER — **Total: 80 Marks**

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